DESCRIPTIVE ANALYTICS

IBM COGNOS _UNION OF TABLES _PROJECT_REPORT PRACTICAL FILE

Submitted by

SHIFA ZAIDI
Roll No: 25050020015

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Mr. Ferhan Ahmed
United University

ABSTRACT

Businesses often store similar information in separate tables. For example, sales transactions for Quarter 1 and Quarter 2 may be maintained in different tables even though both tables contain the same type of information. To analyse the full period, the records can be combined into one dataset.

This project demonstrates the **Union** operation in IBM Cognos Analytics. Union combines rows from two or more compatible data sets into a single result. The project uses two fictional sales tables, Q1_Sales and Q2_Sales, each containing six records and the same six fields. The expected union output contains twelve records.

The report explains the purpose of Union, the difference between Union and Join, input table design, field alignment, workflow construction, execution, expected results, validation, troubleshooting, applications, benefits, and limitations. It also explains that field names and data types should be compatible before the operation is performed.

The key outcome is a practical understanding of how to append records from two similar tables and create one consolidated table for further reporting.

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1. INTRODUCTION

Organisations use data from many sources to prepare reports and make decisions. Sometimes, similar records are stored in separate tables because they belong to different months, quarters, branches, departments, or business units. Analysing each table separately can make it difficult to understand the overall position.

The Union operation helps solve this problem by appending records from compatible tables into one result. For example, a company may have one sales table for the first quarter and another for the second quarter. A Union operation can place all rows into a consolidated sales table.

IBM Cognos Analytics is a business intelligence platform used to prepare, analyse, and present business data. Depending on the Cognos component and version, union functionality may be available through data modules, data preparation, reporting queries, or a supported data modelling workflow. This report uses the general concept of combining compatible query/table outputs with Union.

This project focuses on the row-combination process, not on matching records by key. That distinction is important: Union adds rows, whereas Join connects related rows across tables based on matching fields.

2. OBJECTIVES AND SCOPE

Main objectives

1. To understand the meaning and purpose of the Union operation.
2. To become familiar with the IBM Cognos environment used for data preparation and reporting.
3. To prepare two tables with compatible columns.
4. To combine the two tables into one output using Union.
5. To understand field matching and data-type compatibility.
6. To validate the number of output records and the correctness of values.
7. To identify common errors and practical uses of Union.

Scope

The project uses two fictional quarterly sales tables. Each table contains Order ID, Customer, Region, Category, Sales, and Units. Six rows are provided in each table. The expected combined output contains twelve rows.

Out of scope

The project does not cover complex data warehouse design, ETL automation, security administration, or advanced report visualisation. The precise menu names and available Union workflow depend on the IBM Cognos Analytics version and installed components.

Learning outcome: A student should be able to explain and demonstrate how rows from two structurally compatible tables are appended into one consolidated result.

3. IBM COGNOS OVERVIEW

IBM Cognos Analytics is a business intelligence and reporting platform. It helps organisations connect to data, prepare information, create reports, build dashboards, and share insights. It can work with data from databases, spreadsheets, uploaded files, and enterprise systems, depending on the organisation's setup.

Common capabilities

- Data connectivity: access information from supported databases and files.
- Data preparation: organise and transform data for analysis.
- Data modelling: define relationships and reusable business views.
- Reporting: create tables, charts, and formatted reports.
- Dashboards: present important measures and trends.

Union in a Cognos workflow

Union is a set operation used when two or more input queries or tables have corresponding columns. It appends records from one input to another. Depending on the Cognos version and component, the operation may be configured in a data module, a supported query workflow, or upstream in the database/ETL layer and then consumed by Cognos.

Before starting, confirm that the Cognos environment available to you provides a Union/set-operation feature in the area where you are working. If not, the same union can be prepared in the database or data preparation tool and then imported into Cognos for reporting.

4. UNDERSTANDING THE UNION OPERATION

Union combines records vertically. The columns from each input are aligned, and rows from the second input are added after the rows from the first input in the logical result. The output has one set of columns and contains records from all inputs.

Example

Q1_Sales	Q2_Sales
6 quarterly sales records	6 quarterly sales records
Same 6-column structure	Same 6-column structure

Union versus Join

Union	Join
Adds rows vertically	Combines columns horizontally
Inputs should have compatible column structures	Inputs need a matching key or join condition
Example: Q1 sales + Q2 sales	Example: Sales + Customer details by Customer ID

A standard SQL UNION removes duplicate rows, while UNION ALL retains duplicates. Cognos interfaces may expose different options or behaviours, so verify the selected set operation and duplicate-handling setting in your specific version. This report assumes all twelve example rows are retained.

5. DATASET DESIGN

Two tables are designed to represent sales transactions for two different quarters. They share the same six fields and use compatible data types. This structure makes them suitable for a Union operation.

Field	Data type	Description
Order ID	Integer	Unique transaction identifier
Customer	String	Fictional customer name
Region	String	North, South, East, or West
Category	String	Product group
Sales (■)	Integer / Decimal	Sales value in rupees
Units	Integer	Units sold

Compatibility rules

- Both inputs should contain corresponding fields.
- Corresponding fields should use compatible data types.
- Column order and naming should be checked, especially where matching is positional.
- If one table uses a different column name, map or rename it to the correct output field.
- Missing columns should be added with an appropriate null/default value if required by the tool.

The Order ID values are different across the two tables so that each transaction can be identified in the combined output.

6. INPUT TABLES

Table A: Q1_Sales

Order ID	Customer	Region	Category	Sales (■)	Units
201	Aarav	North	Electronics	12500	12
202	Diya	West	Furniture	8200	8
203	Kabir	South	Electronics	15600	15
204	Meera	East	Stationery	2400	5
205	Ishaan	North	Furniture	9100	9
206	Anaya	West	Stationery	1800	4

Table B: Q2_Sales

Order ID	Customer	Region	Category	Sales (■)	Units
207	Vivaan	South	Electronics	13200	11
208	Sara	East	Furniture	7600	7
209	Reyansh	North	Stationery	3200	6
210	Aditi	West	Electronics	14800	14
211	Arjun	South	Furniture	9900	10
212	Myra	East	Stationery	2100	5

Both tables contain six records and the same six fields. The records represent different transactions. All customer names and values are fictional.

7. PROJECT METHODOLOGY

The project follows a sequence of data preparation, input verification, Union configuration, execution, and output checking.

Step	Activity	Expected result
1	Prepare Q1 and Q2 source tables	Two source datasets ready
2	Connect/import both sources	Both inputs visible in Cognos workflow
3	Check fields and data types	Compatible schemas
4	Add/configure Union operation	Rows appended into one output
5	Run or preview the output	Combined dataset displayed
6	Validate record count and fields	12 correct records

Conceptual workflow

Q1_Sales ■■
■■■ UNION ■■ Combined_Sales
Q2_Sales ■■

The exact node names, canvas layout, and configuration steps vary by Cognos component and version. The diagram represents the logical data flow rather than a screenshot of a particular release.

8. PREPARING THE COGNOS WORKFLOW

1. Open the IBM Cognos Analytics environment provided by your institution or organisation.
2. Open the relevant data preparation, data module, or query-authoring area that supports combining input sets.
3. Import or connect to the first source table, Q1_Sales.
4. Import or connect to the second source table, Q2_Sales.
5. Preview each source to confirm that the six columns and sample values have been read correctly.
6. Check the data types of Order ID, Sales, and Units as numeric, and Customer, Region, and Category as text.
7. Rename or map fields so that the two inputs have matching output names and compatible types.
8. Save the work before configuring the set operation.

Source preparation checklist

- The two input tables are accessible.
- Both tables have six expected fields.
- Field data types are compatible.
- No accidental header row is imported as data.
- Currency values are numeric, not text containing symbols.

Important version note: If your Cognos installation does not provide a Union control in the chosen interface, use a supported data-preparation or database query workflow to create the unioned dataset, then connect that result to Cognos. Do not assume every Cognos screen has an identical Union node.

9. CONFIGURING THE UNION OPERATION

After preparing the two inputs, configure the Union operation in the supported Cognos workflow or underlying query layer.

General configuration procedure

1. Choose the set-operation option labelled Union or the equivalent supported operation.
2. Add Q1_Sales as the first input query/table.
3. Add Q2_Sales as the second input query/table.
4. Match Order ID to Order ID, Customer to Customer, Region to Region, Category to Category, Sales to Sales, and Units to Units.
5. Confirm that the output columns have the intended names and compatible data types.
6. Decide whether duplicate rows should be removed or retained. For this example, retain all rows (the equivalent of UNION ALL where available), because the goal is to append every transaction.
7. Save the configuration.

Expected output schema

Order ID	Customer	Region	Category	Sales (■)	Units
Integer	Text	Text	Text	Numeric	Integer

The Union output contains one set of six columns. The rows from both input tables are appended into that output.

10. RUNNING AND VIEWING THE OUTPUT

1. Save the Cognos work or data preparation flow.
2. Run, refresh, or preview the combined query according to the available interface.
3. Open the output preview or create a simple list/table report using the unioned result.
4. Check the first few records from Q1_Sales and confirm that records from Q2_Sales are also present.
5. Confirm that the output contains twelve records when duplicates are retained and no filters are applied.
6. Save the final output or report for future use.

Expected result

The output should contain all six Q1 records and all six Q2 records. Each record should retain its original Order ID, Customer, Region, Category, Sales, and Units values.

Suggested screenshots for your practical file

(a) Q1_Sales input preview, (b) Q2_Sales input preview, (c) workflow showing both inputs connected to Union, (d) Union field mapping/settings, and (e) final combined output table.

Add screenshots from your own Cognos session. The screenshots should show the actual version and interface used by your institution.

11. RESULTS AND INTERPRETATION

The following table represents the expected output when the two six-row input tables are combined and duplicate rows are retained.

Order ID	Customer	Region	Category	Sales (■)	Units
201	Aarav	North	Electronics	12500	12
202	Diya	West	Furniture	8200	8
203	Kabir	South	Electronics	15600	15
204	Meera	East	Stationery	2400	5
205	Ishaan	North	Furniture	9100	9
206	Anaya	West	Stationery	1800	4
207	Vivaan	South	Electronics	13200	11
208	Sara	East	Furniture	7600	7
209	Reyansh	North	Stationery	3200	6
210	Aditi	West	Electronics	14800	14
211	Arjun	South	Furniture	9900	10
212	Myra	East	Stationery	2100	5

Result summary

Measure	Value
Rows in Q1_Sales	6
Rows in Q2_Sales	6
Expected rows after append	12
Output columns	6

The output includes transactions from both quarters. The number of rows is the sum of the input row counts because this example retains duplicates and applies no filters. In a real Union operation that removes duplicates, the final count can be lower if identical rows exist in both inputs.

12. VALIDATION AND TROUBLESHOOTING

Validation checklist

- Confirm that Q1_Sales contains six records and Q2_Sales contains six records.
- Confirm that the combined output contains twelve records when duplicates are retained.
- Confirm that all six expected columns are present.
- Check that Order IDs 201 to 212 each appear once in this sample.
- Compare sample values with the original input tables.
- Confirm that Sales is numeric and can be aggregated correctly.

Common problems and solutions

Problem	Possible cause	Solution
Fields do not align	Different names/order	Rename or map corresponding fields
Type mismatch	Text vs numeric	Convert fields to compatible types
Fewer rows than expected	Duplicate removal or filter	Check Union mode and upstream filters
Missing output values	A field absent in one input	Add/map a suitable null or default field
Repeated records	Duplicates retained	Use duplicate-removal Union if appropriate

If the output count differs from twelve, first check whether the selected operation removes duplicates, whether a filter is applied, and whether both inputs were included.

13. APPLICATIONS, BENEFITS AND LIMITATIONS

Applications

Union is useful for combining monthly or quarterly sales tables, merging branch-wise transaction files, appending survey responses collected in separate batches, consolidating attendance records, and combining historical and current data that share the same structure.

Benefits

- Creates one consolidated dataset from multiple similar sources.
- Makes period-wise reporting easier.
- Reduces the need to analyse each source separately.
- Supports consistent downstream reporting and visualisation.
- Can be repeated when new batches of compatible data arrive.

Limitations and precautions

- Inputs must have compatible structures and data types.
- A Union does not match records by a key or enrich a row with columns from another table.
- Duplicate handling must be understood: duplicate removal can change row counts.
- Inconsistent field names, formats, or units can produce incorrect output.
- Large unions may require performance planning and appropriate database processing.

Union should be selected when the goal is to append records. If the goal is to combine customer details with sales transactions using Customer ID, a Join is more suitable.

14. CONCLUSION AND REFERENCES

Conclusion

This project explained the Union operation in IBM Cognos using two fictional quarterly sales tables. Each input table contained six rows and the same six fields. After combining the tables while retaining duplicates, the expected output contained twelve rows and one common set of six columns.

The project highlighted the importance of compatible field names, data types, and correct field mapping. It also explained how Union differs from Join: Union appends rows vertically, while Join combines related columns horizontally based on a matching condition.

The main learning is that Union is useful for consolidating similar datasets from different periods or sources. Before using the result for reporting, users should validate row counts, field values, data types, and duplicate-handling behaviour.

References

1. IBM Documentation, IBM Cognos Analytics product documentation: https://www.ibm.com/docs/en/cognos-analytics
2. IBM Cognos Analytics installed product help and training materials (version-specific guidance).
3. Project dataset: fictional Q1 and Q2 sales records created specifically for this report.

Appendix: screenshots to add before submission

Insert screenshots of the two source tables, the Union configuration or set-operation query, and the final combined output from your own IBM Cognos environment. This will make the report evidence-based and specific to the version used.

End of Report